

Curriculum Vitae

Layas Thapaliya

Graduate student in Civil and Environmental Engineering with a strong academic foundation in mass transfer mechanisms within bioprocesses. Interested in exploring how nanobubble technology can be applied to optimize system efficiency and support resource recovery initiatives. Interested to further develop technical skills in physical-chemical applications and contribute to sustainable engineering solutions within the field of environmental biotechnology.

Professional Experience

● Feb 2026–Current

Graduate Research Assistant

Department of Civil and Environmental Engineering, HKUST

- Research focus: HOB system optimization
- PI: Prof. Samir Kumar Khanal

● Apr 2024–Sep 2025

Undergraduate Research Assistant

Department of Applied Sciences and Chemical Engineering, Pulchowk Campus, TU, Nepal

- Research Focus: Bromelain extraction from pineapple peels by fermentation
- PI: Dr. Sahira Joshi & Dr. Nirmal Ghimire

● Nov 2023–Dec 2023

Undergraduate Research Intern

Riddhi Siddhi Cement, Hetauda, Nepal

- Role: Quality Control and testing

Research Areas

- Gas Fermentation
- Nanobubble Technology
- Nutrient and Resource Recovery



Contact



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The Hong Kong University of Science and Technology
Clear Water Bay, Kowloon
Hong Kong



[Layas Thapaliya](#)

Education

MPhil in Civil and Environmental Engineering (Currently)

(Expected Feb 2028)

The Hong Kong University of Science and Technology

B.E. in Chemical Engineering

(Apr 2025)

Tribhuvan University, Nepal

GPA: 81.02/100

Skills & Expertise

- R-programming, C-programming, dart programming
- Web Development: HTML, CSS, JavaScript
- Solvent and Enzyme Extraction
- Calibrating and operating UV-vis spectrophotometer

Awards and Honors

1. **PostGraduate Studentship**–The Hong Kong University of Science and Technology – (Feb 2026–Current)
2. **Merit Scholarship (\$4000)**–Tribhuvan University – (Apr 2021–April 2025)

Certifications

1. Responsible Conduct of Research (1-RCR), The Hong Kong University of Science and Technology, Apr 2026
2. Lab Safety Training on Chemical, Biological and Fire Safety, Feb 2026

Relevant Courses

Graduate Courses (MPhil in Civil and Environmental Engineering)

- Resource Recovery from Bioresources (CIVL4100P)
- Aquatic Chemistry (CIVL 5430)

Undergraduate Courses (B.E. in Chemical Engineering)

- Transport Phenomena (EC756)
- Biochemical Engineering (EC752)
- Maintenance Engineering and Safety (EC707)
- Modelling and Simulation in Chemical Engineering (EC706)
- Chemical Engineering Design I/II (EC651/EC701)
- Mass Transfer I/II (EC602/EC654)
- Chemical Reaction Engineering I/II (EC606/EC655)
- Process Economics and Plant Design (EC708)
- Chemical Process Calculation (EC504)
- Computer Aided Drawing (ME454)
- Chemical Engineering Thermodynamics I/II (EC551/EC601)
- Mechanical Operation (EC604)
- Instrumentation and Process Control (EC652)
- Chemical Process Industries I/II (EC552/EC603)